

A113 · A Novel ILP Framework to Identify Compensatory Pathways in Genetic Interaction Networks with GIDEON

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In Baker's yeast, there exists a comprehensive collection of pairwise epistasis experiments that, for nearly every pair of non-essential genes, measure the growth of the double-knockout strain as compared to its component single knockouts. This data can be represented as a weighted signed graph termed the genetic interaction network, and we introduce a new ILP-based method named GIDEON to search for a diverse collection of Between-Pathway Models (BPMs) in this network, where BPMs are a graph motif signature that indicates potential compensatory pathways in the genetic interaction network.

With both an improved distribution-informed edge weighting scheme and an improved ILP method, GIDEON produces BPM collections that are substantially larger and with better functional enrichment compared to previous methods. We find some interesting new BPM gene sets including one with potential insights into antifungal drug targets through ties between ergosterol and aromatic amino acid biosynthesis.